

Document 04: Functional Requirements Specification (FRS)

Project: Natural Intellects School Management System (NISMS)

Version: 1.0

Status: Master Functional Requirements Document

Purpose:

This document defines all functional requirements, user permissions, workflows, business rules, validations, and operational requirements for the Natural Intellects School Management System.

It serves as the primary implementation reference for:

- Figma design work
- Database design
- Backend development
- Frontend development
- Cloud AI implementation
- Testing
- Future maintenance

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1. USER ROLES & PERMISSIONS

Super Administrator

Permissions:

- Full system access
 - User management
 - School configuration
 - System settings
 - Backups
 - Security settings
-

Proprietor

Permissions:

- Financial reports
- Student statistics
- School dashboards
- Administrative oversight

Restrictions:

- No technical configuration permissions.
-

Head Teacher

Permissions:

- Teacher supervision
- Academic reports
- Attendance reports
- Student oversight

School Secretary

Permissions:

- Student registration
- Teacher registration
- Parent management
- Printing records

Bursar

Permissions:

- Fee management
- Payment recording
- Receipt generation
- Financial reports

Restrictions:

- No academic permissions.

Teacher

Permissions:

- Assigned classes
- Attendance submission
- Marks entry
- Student viewing

2. PUBLIC WEBSITE MODULE

Purpose

Provide schools with a professional digital presence.

Public Pages

The website shall include:

Home Page

Features:

- School introduction
 - Hero section
 - News highlights
 - Call to action
-

About Page

Features:

- School history
 - Mission
 - Vision
 - Leadership
-

Admissions Page

Features:

- Admission procedures
 - Requirements
 - Downloadable forms
-

Academics Page

Features:

- Programs
 - Departments
 - Subjects
 - Performance information
-

Gallery

Features:

- School images
 - Events
 - Activities
-

News & Events

Features:

- Announcements
 - Events
 - Updates
-

Contact Page

Features:

- Phone numbers
 - Email addresses
 - Maps
 - Contact forms
-

3. STUDENT MANAGEMENT MODULE

Functional Requirements

The system shall support:

Student Registration

Required Fields:

Personal:

- First Name
- Middle Name
- Last Name
- Gender
- Date of Birth
- Photo

School Information:

- Admission Number
- Admission Date
- Class
- Stream

Status:

- Active
 - Graduated
 - Transferred
 - Archived
-

Student Search

Search By:

- Name
 - Admission Number
 - Class
 - Stream
 - Guardian
-

Student Profiles

Profiles shall display:

- Personal information
 - Guardian information
 - Attendance summaries
 - Fee summaries
 - Academic summaries
-

Student Promotion

Support:

- Individual promotion
 - Bulk promotion
-

Student Transfers

Support:

- Internal transfers
 - External transfers
-

Student Archiving

Deletion shall not be permitted.

Historical records must remain intact.

4. TEACHER MANAGEMENT MODULE

Functional Requirements

The system shall support:

Teacher Registration

Required Information:

Personal:

- Names
- Gender
- DOB
- Photo

Contacts:

- Phone
- Address
- Email

Employment:

- Staff Number
 - Position
 - Employment Date
-

Subject Assignment

Support:

One Teacher:

↓

Many Subjects

Class Assignment

Support:

One Teacher:

↓

Multiple Classes

Teacher Profiles

Include:

- Personal information
 - Subjects
 - Assigned classes
 - Employment information
-

5. CLASS & STREAM MANAGEMENT MODULE

Functional Requirements

Class Management

Examples:

Primary:

- P1
- P2
- P3

Secondary:

- S1
- S2
- S3

Schools shall customize classes.

Stream Management

Examples:

P4 Blue

P4 Red

S2 East

S2 West

Class Teacher Assignment

One class may have:

- One class teacher

Multiple subject teachers.

6. SUBJECT MANAGEMENT MODULE

Functional Requirements

Subject Creation

Examples:

Primary:

- English
- Mathematics
- SST
- Science

Secondary:

- Physics
 - Chemistry
 - Biology
-

Subject Assignment

Support:

Teacher:



Subjects



Classes

Subject Categories

Examples:

Sciences

Arts

Languages

Optional support.

7. ATTENDANCE MANAGEMENT MODULE

Teacher Functions

Teachers shall:

- Select classes
 - Mark attendance
 - Submit records
-

Attendance States

Students may be:

- Present
- Absent
- Sick

- Excused
-

Administrative Functions

Reports:

- Daily attendance
 - Weekly attendance
 - Monthly attendance
-

Business Rules

Attendance shall only be recorded once per day per student.

8. ACADEMIC REPORTS MODULE

Marks Entry

Teachers shall:

- Select classes
 - Select subjects
 - Enter marks
-

Grading

Support:

- Percentage grading
- Custom grading systems

Examples:

A

B

C

D

F

Terms

Support:

- Term I
 - Term II
 - Term III
-

Report Generation

The system shall support:

Printable report cards.

9. FEE MANAGEMENT MODULE

Payment Methods

Support:

- Cash
 - Mobile Money
 - Bank Deposits
-

Student Accounts

Students shall have:

- Current balances
 - Payment histories
 - Receipts
-

Receipt Generation

Receipts shall include:

- Student names
- Payment amounts

- Dates
 - References
-

Business Rules

Payments cannot be deleted after confirmation.

Corrections shall use adjustment mechanisms.

10. PARENT & GUARDIAN MANAGEMENT MODULE

Multiple Guardians

Support:

Examples:

- Father
 - Mother
 - Sponsor
 - Relative
-

Guardian Information

Required:

- Names
 - Phone numbers
 - Addresses
-

Emergency Contacts

Support:

Emergency guardian information.

11. ANNOUNCEMENT MODULE

School Announcements

Support:

- General announcements
 - Event notices
 - Printable notices
-

Class Announcements

Support:

Class-specific communications.

Future Features

Potential support:

- SMS notifications
 - WhatsApp integrations
-

12. GLOBAL BUSINESS RULES

Student Rules

Students:

- Cannot be permanently deleted.
 - Must belong to one class.
 - May have multiple guardians.
-

Teacher Rules

Teachers:

- Cannot be permanently deleted.
- May teach multiple subjects.

- May teach multiple classes.
-

Attendance Rules

Attendance:

- One record per student per day.
-

Financial Rules

Payments:

- Cannot be deleted.
 - Must maintain historical integrity.
-

13. VALIDATION RULES

Names

Maximum:

100 characters.

Phone Numbers

Support:

070xxxxxxx

+25670xxxxxxx

Images

Allowed:

- JPG
 - PNG
 - WEBP
-

Dates

Future dates shall not be allowed where inappropriate.

14. REPORTING REQUIREMENTS

The system shall support:

Student Reports

Examples:

- Enrollment reports
 - Gender reports
 - Class reports
-

Teacher Reports

Examples:

- Teacher workloads
 - Subject allocations
-

Attendance Reports

Examples:

- Daily summaries
 - Monthly reports
-

Academic Reports

Examples:

- Performance reports
 - Term reports
-

Financial Reports

Examples:

- Outstanding balances
 - Collection summaries
-

15. FUTURE CONSIDERATIONS

Potential future modules:

- Parent portals
- SMS integrations
- Library systems
- Inventory systems
- Hostel systems
- Online applications

These modules remain outside the current implementation scope.

STRATEGIC CONCLUSIONS

The Functional Requirements Specification shall serve as the authoritative implementation document for:

- Figma designs
- Database architecture
- Backend systems
- Frontend interfaces
- Cloud AI development workflows

All future implementations shall comply with the requirements defined within this document.

End of Document 04.